

CSCE 763: Digital Image Processing

Homework #1

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University of South Carolina
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Due: 1:15 pm EST, Tuesday, Feb 3

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Problem 1: Smallest Discernible Dot (25 pts)

Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.5 m away from the eyes.

Assumptions:

1. The distance between the center of the lens and the retina along the visual axis is 14 mm.
2. The visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina.
3. The fovea can be modeled as a square array of dimensions $1.5 \text{ mm} \times 1.5 \text{ mm}$, and the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Step 1: Determine the Diameter of a Single Cone

The fovea is a $1.5 \text{ mm} \times 1.5 \text{ mm}$ square with 337,000 cones uniformly distributed. Modeling as a square grid, the number of cones per side is:

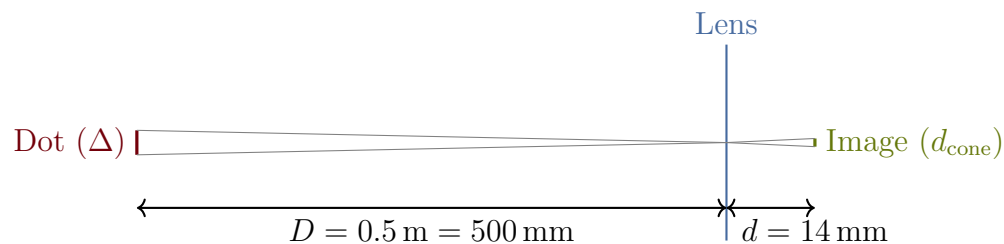
$$n = \sqrt{337,000} \approx 580.5$$

The center-to-center spacing (cone diameter including gaps) is:

$$d_{\text{cone}} = \frac{1.5 \text{ mm}}{580.5} \approx 0.00258 \text{ mm} = 2.58 \mu\text{m}$$

Step 2: Apply Similar Triangles

The dot on the page and its image on the retina are related by similar triangles through the lens center:



By similar triangles:

$$\frac{\Delta}{D} = \frac{d_{\text{cone}}}{d}$$

Step 3: Solve for Minimum Dot Diameter

The smallest detectable image equals one cone diameter: $d_{\text{image}} = d_{\text{cone}} \approx 0.00258 \text{ mm}$.

$$\Delta = D \cdot \frac{d_{\text{cone}}}{d} = 500 \cdot \frac{0.00258}{14} = \frac{1.29}{14} \approx 0.0922 \text{ mm}$$

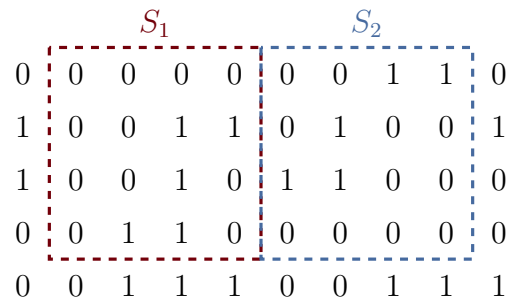
Results

The diameter of the smallest printed dot that the eye can discern at a distance of 0.5 m is:

$$\Delta \approx 0.0922 \text{ mm} \approx 92.2 \mu\text{m}$$

Problem 2: Adjacency of Image Subsets (25 pts)

Consider the two image subsets S_1 and S_2 shown below. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.



Step 1: Identify Pixels in V

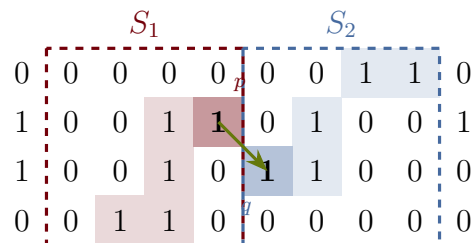
The subsets share the boundary between columns 4 (rightmost of S_1) and 5 (leftmost of S_2). Using (x, y) coordinates from the grid:

S_1 pixels with value 1: $(3, 1)$, $(4, 1)$, $(3, 2)$, $(2, 3)$, $(3, 3)$.

S_2 pixels with value 1: $(7, 0)$, $(8, 0)$, $(6, 1)$, $(5, 2)$, $(6, 2)$.

Step 2: Identify the Closest Cross-Boundary Pair

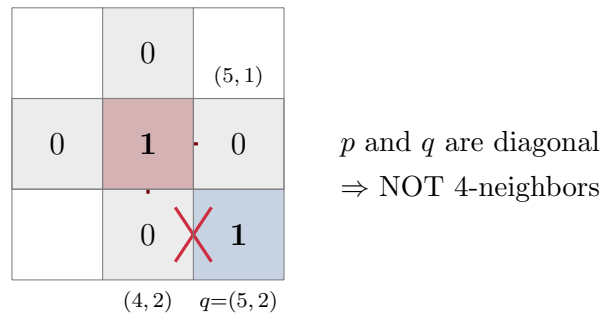
The closest cross-boundary pair in V is $p = (4, 1) \in S_1$ and $q = (5, 2) \in S_2$.



Shaded cells have value $1 \in V$. The darkened cells are the closest cross-boundary pair $p = (4, 1)$ and $q = (5, 2)$.

Part (a): 4-Adjacency

Two subsets are 4-adjacent if $\exists p \in S_1, q \in S_2$ with values in V that are 4-neighbors (share an edge). Two pixels are 4-neighbors if they differ in exactly one coordinate by 1.

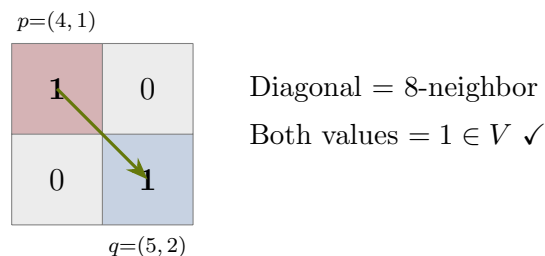


The nearest cross-boundary pair in V is $(4,1)$ and $(5,2)$, which differ in *both* coordinates — they are *not* 4-neighbors. No other cross-boundary pair of pixels in V is closer.

$\Rightarrow S_1$ and S_2 are NOT 4-adjacent.

Part (b): 8-Adjacency

Two pixels are 8-neighbors if they differ by at most 1 in each coordinate (including diagonals).



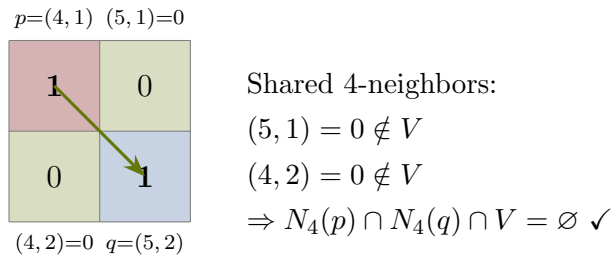
$(4,1)$ and $(5,2)$ differ by 1 in each coordinate, so they are diagonal 8-neighbors. Both have value $1 \in V$.

$\Rightarrow S_1$ and S_2 ARE 8-adjacent.

Part (c): m-Adjacency

Two pixels p, q are m-adjacent if:

- They are 4-adjacent with values in V , **or**
- They are diagonal 8-neighbors with values in V **and** $N_4(p) \cap N_4(q) \cap V = \emptyset$ (no shared 4-neighbor has a value in V).



Since $N_4(p) \cap N_4(q) \cap V = \emptyset$, the m-adjacency condition is satisfied.
 $\Rightarrow S_1$ **and** S_2 **ARE m-adjacent.**

Results

For $V = \{1\}$:

- S_1 and S_2 are **not** 4-adjacent.
- S_1 and S_2 **are** 8-adjacent (via $(4,1) \leftrightarrow (5,2)$).
- S_1 and S_2 **are** m-adjacent (via $(4,1) \leftrightarrow (5,2)$, since $N_4(p) \cap N_4(q) \cap V = \emptyset$).

Problem 3: Shortest Paths (25 pts)

Consider the image segment shown below.

- (a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m-path between p and q . If a particular path does not exist between these two points, explain why.
- (b) Repeat for $V = \{1, 2\}$.

(q)

3	1	2	1
2	0	0	2
1	2	1	1
1	0	1	2

(p)

Problem 3(a): $V = \{0, 1\}$ **Step 1: Identify Pixels in V**

Using (c, r) (column, row) coordinates with the origin at the top-left:

3	1	2	1 ^{q}
2	0	0	2
1	2	1	1
1 _{p}	0	1	2

Shaded cells have values in V . $p = (0, 3) = 1 \in V$, $q = (3, 0) = 1 \in V$.

Step 2: Shortest 4-Path

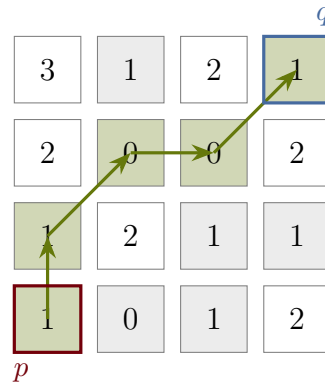
For a 4-path, each step moves to a horizontal or vertical neighbor in V .

3	1	X	1 ^{q}	q 's 4-neighbors: $(2, 0) = 2 \notin V$ $(3, 1) = 2 \notin V$ q is 4-isolated
2	0	0	X	
1	2	1	1	
1 _{p}	0	1	2	

$\Rightarrow$ No 4-path exists.

Step 3: Shortest 8-Path

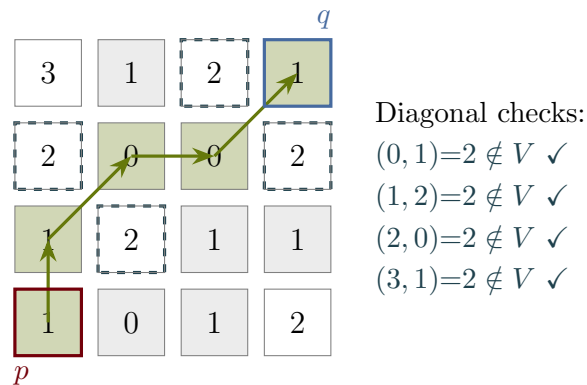
For an 8-path, each step moves to any of the 8 surrounding neighbors in V .



Path: $(0, 3) \rightarrow (0, 2) \rightarrow (1, 1) \rightarrow (2, 1) \rightarrow (3, 0)$. Length = 4.

Step 4: Shortest m-Path

For m-adjacency, a diagonal step is allowed only if no shared 4-neighbor has a value in V .

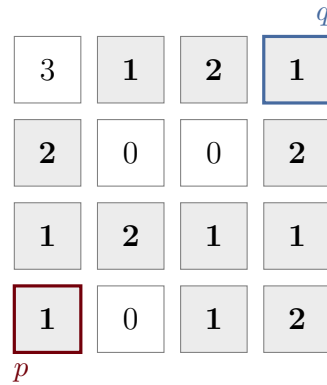


All diagonal steps satisfy m-adjacency (dashed cells are shared 4-neighbors, all $\notin V$). Same path is valid. Length = 4.

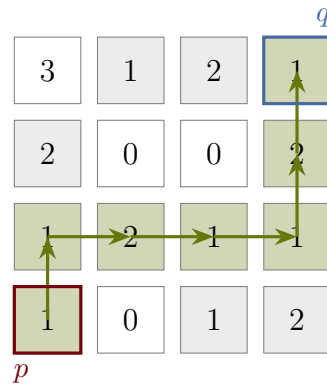
Results

For $V = \{0, 1\}$:

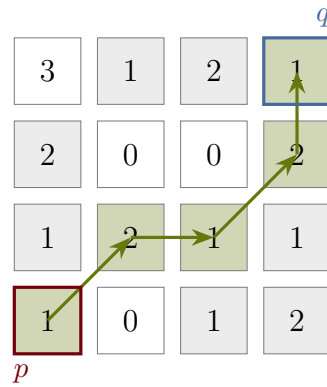
- **Shortest 4-path:** Does not exist (q is 4-isolated in V).
- **Shortest 8-path:** Length 4: $(0, 3) \rightarrow (0, 2) \rightarrow (1, 1) \rightarrow (2, 1) \rightarrow (3, 0)$.
- **Shortest m-path:** Length 4: same path (all diagonals satisfy m-adjacency).

Problem 3(b): $V = \{1, 2\}$ **Step 1: Identify Pixels in V** 

Shaded cells have values in V . Both p and q are well-connected.

Step 2: Shortest 4-Path

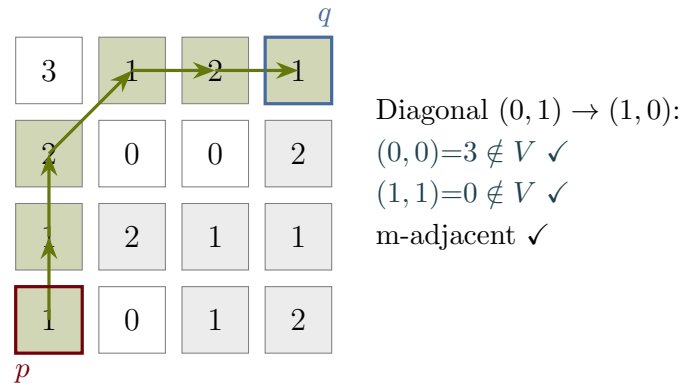
Path: $(0, 3) \rightarrow (0, 2) \rightarrow (1, 2) \rightarrow (2, 2) \rightarrow (3, 2) \rightarrow (3, 1) \rightarrow (3, 0)$. Length = **6**. All 4-adjacent steps in V . No shorter 4-path exists because $(1, 1)$ and $(2, 1)$ have value $0 \notin V$, blocking direct routes.

Step 3: Shortest 8-Path

Path: $(0,3) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (3,1) \rightarrow (3,0)$. Length = 4.

Step 4: Shortest m-Path

The diagonals in the 8-path both fail m-adjacency (shared 4-neighbors in V), so we need an alternative route. Going up the left column and across the top row allows a single valid diagonal:



Path: $(0,3) \rightarrow (0,2) \rightarrow (0,1) \rightarrow (1,0) \rightarrow (2,0) \rightarrow (3,0)$.

- $(0,3) \rightarrow (0,2)$: 4-adj, values $1, 1 \in V \checkmark$
- $(0,2) \rightarrow (0,1)$: 4-adj, values $1, 2 \in V \checkmark$
- $(0,1) \rightarrow (1,0)$: diagonal, values $2, 1 \in V$. Shared 4-neighbors: $(0,0) = 3 \notin V$, $(1,1) = 0 \notin V$. m-adjacent $\checkmark$
- $(1,0) \rightarrow (2,0)$: 4-adj, values $1, 2 \in V \checkmark$
- $(2,0) \rightarrow (3,0)$: 4-adj, values $2, 1 \in V \checkmark$

Length = 5.

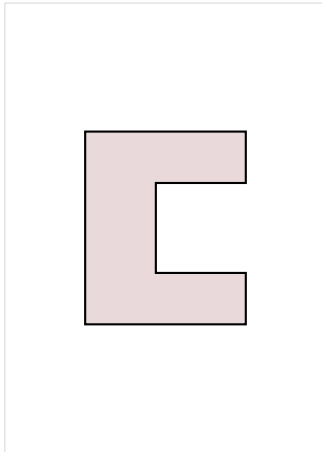
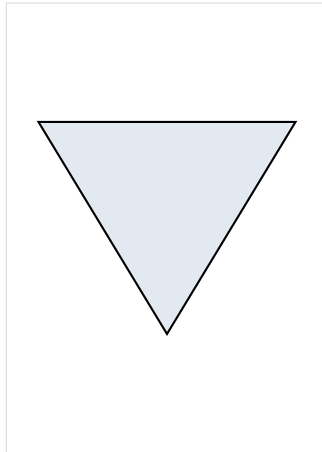
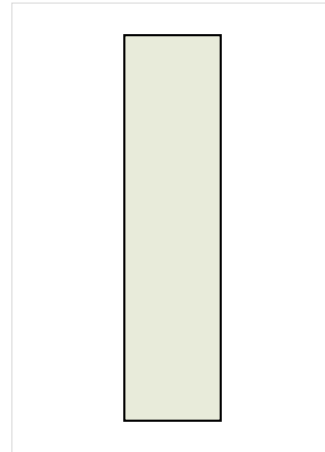
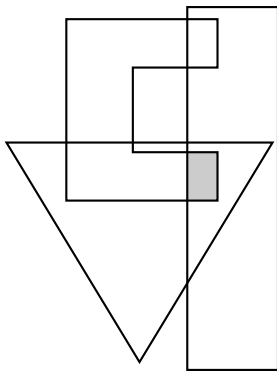
Results

For $V = \{1, 2\}$:

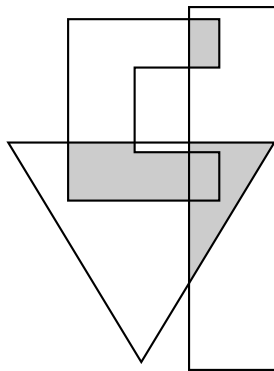
- **Shortest 4-path:** Length 6: $(0,3) \rightarrow (0,2) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,1) \rightarrow (3,0)$.
- **Shortest 8-path:** Length 4: $(0,3) \rightarrow (1,2) \rightarrow (2,2) \rightarrow (3,1) \rightarrow (3,0)$.
- **Shortest m-path:** Length 5: $(0,3) \rightarrow (0,2) \rightarrow (0,1) \rightarrow (1,0) \rightarrow (2,0) \rightarrow (3,0)$.

Problem 4: Set Expressions (25 pts)

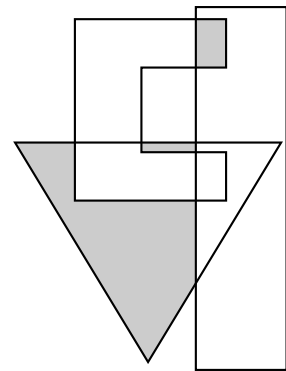
Give expressions for the sets shown shaded in the following figure in terms of sets A , B , and C . The shaded areas in each figure constitute one set, so give one expression for each of the three figures.

 A  B  C 

(i)



(ii)



(iii)

Figure (i)

We systematically check each possible region of the Venn diagram to determine which are shaded:

Region	Set Expression	Shaded?
A only	$A \cap B' \cap C'$	No
B only	$B \cap A' \cap C'$	No
C only	$C \cap A' \cap B'$	No
$A \cap B$ only	$A \cap B \cap C'$	No
$A \cap C$ only	$A \cap C \cap B'$	No
$B \cap C$ only	$B \cap C \cap A'$	No
$A \cap B \cap C$	$A \cap B \cap C$	Yes

The only shaded region is where all three sets overlap simultaneously. Therefore:

$$\boxed{A \cap B \cap C}$$

Figure (ii)

Checking each region of the Venn diagram:

Region	Set Expression	Shaded?
A only	$A \cap B' \cap C'$	No
B only	$B \cap A' \cap C'$	No
C only	$C \cap A' \cap B'$	No
$A \cap B$ only	$A \cap B \cap C'$	Yes
$A \cap C$ only	$A \cap C \cap B'$	Yes
$B \cap C$ only	$B \cap C \cap A'$	Yes
$A \cap B \cap C$	$A \cap B \cap C$	Yes

Every region where at least two sets overlap is shaded. We can write the full expression as:

$$(A \cap B \cap C') \cup (A \cap C \cap B') \cup (B \cap C \cap A') \cup (A \cap B \cap C)$$

This simplifies. Notice that $(A \cap B \cap C') \cup (A \cap B \cap C) = A \cap B$, and similarly for the other pairs. Collecting terms:

$$\boxed{(A \cap B) \cup (B \cap C) \cup (A \cap C)}$$

Figure (iii)

Checking each region of the Venn diagram:

Region	Set Expression	Shaded?
A only	$A \cap B' \cap C'$	No
B only	$B \cap A' \cap C'$	Yes
C only	$C \cap A' \cap B'$	No
$A \cap B$ only	$A \cap B \cap C'$	No
$A \cap C$ only	$A \cap C \cap B'$	Yes
$B \cap C$ only	$B \cap C \cap A'$	No
$A \cap B \cap C$	$A \cap B \cap C$	No

Two disjoint regions are shaded. Taking the union of their set expressions directly from the table:

$$(B \cap A' \cap C') \cup (A \cap C \cap B')$$

These two regions share no overlap, so no further simplification is possible:

$$\boxed{(B \cap A' \cap C') \cup (A \cap C \cap B')}$$

Results

(i) $A \cap B \cap C$

(ii) $(A \cap B) \cup (B \cap C) \cup (A \cap C)$

(iii) $(B \cap A' \cap C') \cup (A \cap C \cap B')$